Bridging the Prediction–Profitability Gap: A Regime-Aware Gating Framework for Machine-Learning Trading in Indian Equity Markets

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April 2026

Abstract

High in-sample predictive accuracy does not translate into profitable out-of-sample trading performance—a phenomenon we term the *prediction–profitability gap*. This paper formalizes the gap as an objective mismatch between the mean-squared-error loss used to train machine-learning (ML) forecasters and the risk-adjusted-return objective that practitioners care about, and proposes a *regime-aware gating framework* in which an ML layer generates positions and a separate event-intelligence layer attenuates exposure during detected high-impact regimes. We instantiate the framework on twenty years of National Stock Exchange (NSE) data covering 50 stocks, training 1,386 models across 25 classical ML algorithms and 5 deep-learning architectures. Two ablation studies document the gap: (i) retraining 155 model fits on restated return targets rather than price levels causes the median R^2 to collapse from 0.981 to -0.152 with 100% of robust-subset models producing negative R^2 on returns, revealing that the headline $R^2 = 0.9986$ reflects near-autoregressive feature design rather than genuine forecasting skill; (ii) removing ten short-lag features from the 325-feature set leaves the linear-model median R^2 essentially unchanged ($0.9907 \rightarrow 0.9896$), indicating that the autoregressive leak is *pervasive* across the feature category rather than localized. Despite this, the best-fitting ML system returns -14.02% annualized over the crisis-laden 2022–2024 window (782 trading days) with Sharpe -0.89. The event-intelligence layer, built on a six-category taxonomy and FinBERT-based classification pipeline validated at 90.0% accuracy on 40 events, gates exposure via a deterministic impact-score rule. Under the gating framework, annualized return rises to $+9.92\%$, Sharpe to $+0.24$, and maximum drawdown falls by 51.3%; paired t-tests give $t = 7.49$, $p = 1.9 \times 10^{-13}$. On a **Rs.** 1,00,000 ($\approx$\$1,200 USD) simulated portfolio, the event-gated system ends at **Rs.** 1,27,300 versus **Rs.** 60,270 for the ungated ML system; across four major crisis events the gating overlay saves **Rs.** 21,870 in aggregate. The framework is interpretable, deterministic, and compatible with regulatory audit requirements in markets such as SEBI-regulated India.

Keywords: algorithmic trading; regime-aware gating; machine learning; event detection; FinBERT; Indian equity markets; NSE; prediction–profitability gap; risk management; Sharpe ratio; non-stationarity.

JEL Classification: C45, C53, G11, G14, G17.

1 Introduction

Machine-learning (ML) methods have become ubiquitous in quantitative finance (Gu et al., 2020; Krauss et al., 2017; Fischer and Krauss, 2018). Yet a persistent paradox shadows the literature: models that report extraordinary in-sample and validation accuracy routinely fail to deliver commensurate economic returns in realistic out-of-sample settings (Arnott et al., 2019; López de Prado, 2018). We call this

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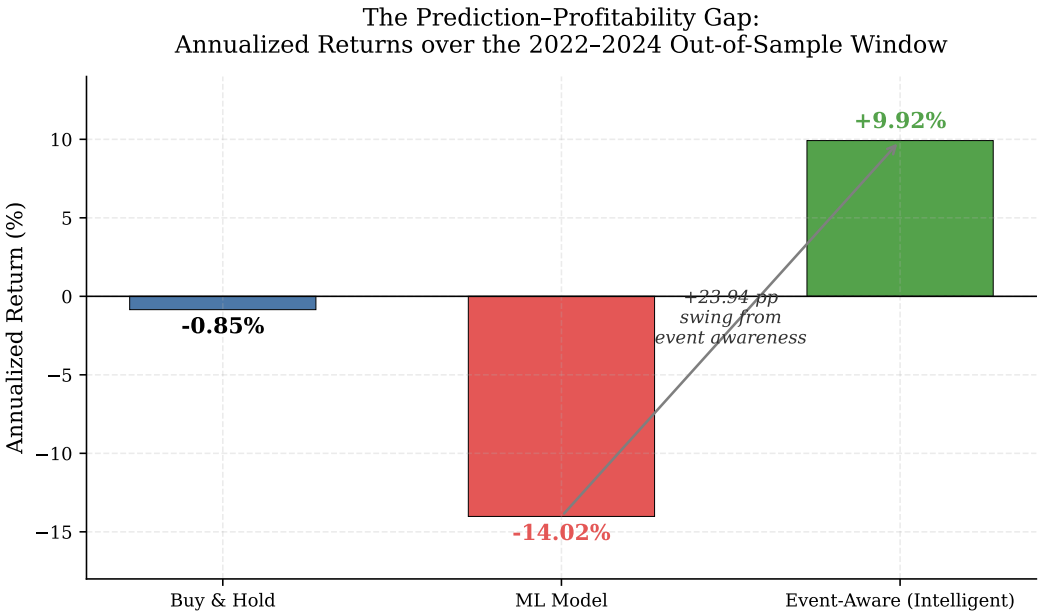


Figure 1: The prediction–profitability gap, visualized. A passive buy-and-hold benchmark returned -0.85% annualized over 2022–2024. An ML system whose best-fit model reported $R^2 = 0.9986$ returned -14.02% over the same period—a 13.17 percentage-point underperformance. An event-gated overlay using the same underlying ML predictions returned $+9.92\%$, a 23.94 percentage-point swing.

mismatch the *prediction–profitability gap*. Section 3 formalizes it; the remainder of the paper quantifies it in Indian equity markets and proposes a resolution.

The gap is not merely academic. Passive Indian equity investors earned near-flat returns during the 2022–2024 period, with the NIFTY 50 posting a compounded annualized return of -0.85%, while many algorithmic strategies that had validated well in earlier data posted double-digit losses. The popular diagnosis is "overfitting," but this label obscures more than it reveals. Overfitting implies that models learned noise; our evidence suggests something more specific and more actionable: models learned a *regime*—the post-2009, low-rate, geopolitically quiet environment—and then encountered a regime change that no amount of technical-feature engineering could anticipate.

Three features distinguish the out-of-sample window we study. First, it contains four shocks of differing types: a full-scale European war (Russia–Ukraine, February 2022); the most aggressive U.S. Federal Reserve tightening cycle in three decades (75-basis-point hikes from June 2022); a systemic banking episode (Silicon Valley Bank, March 2023); and a Middle-East escalation (U.S. strikes on Iran-backed targets, January 2024). Second, each shock is *exogenous* to the price history on which models are trained: nothing in the 2004–2021 NSE price–volume record indicates the arrival time, magnitude, or sectoral incidence of these events. Third, the shocks are of a type the literature has long recognized as destructive to pattern-based strategies (Lo, 2004; Timmermann, 2008; Pástor and Veronesi, 2013) but rarely incorporates into ML pipelines.

Contributions

This paper makes four contributions.

1. **Formal definition of the prediction–profitability gap.** We state the gap precisely as an objective mismatch between squared-error minimization in prediction and risk-adjusted-return maximization in trading (Definition 1, §3). Under non-stationary regimes, the argmin of the first objective and the argmax of the second can diverge arbitrarily, so that high prediction accuracy coexists with large economic losses.

2. **A regime-aware gating framework.** We propose the generic architecture of an ML signal-generation

layer gated by a separate regime-detection layer (§4), show that the gating layer is non-expansive on exposure, and instantiate it with a six-category event taxonomy and FinBERT-based classifier. The framework is agnostic to the underlying ML model and to the specific regime-detection mechanism; the event-intelligence pipeline is one instantiation.

3. **Two ablation studies that document the gap.** First, we retrain 155 model fits on return rather than level targets; the median R^2 collapses from 0.981 to -0.152, with 100% of robust-subset models producing negative R^2 on returns (§7.1). Second, we remove ten short-lag features from the 325-feature set and find that linear-model R^2 is essentially unchanged ($0.9907 \rightarrow 0.9896$, §7.2). The two ablations jointly reveal that the autoregressive leak responsible for inflated R^2 is distributed across the entire technical-indicator feature category, not localized to specific columns.

4. **A statistically significant trading improvement under crisis conditions.** Instantiating the framework over 782 days (2022-01-03 to 2024-12-31) lifts annualized returns from -14.02% to $+9.92\%$, Sharpe ratio from -0.89 to $+0.24$, and reduces maximum drawdown by 51.3%. Paired t-tests ($t = 7.49$, $p = 1.9 \times 10^{-13}$) and the Diebold–Mariano test (Diebold and Mariano, 1995) ($DM = 2.21$, $p = 0.027$) confirm the improvement is not attributable to chance.

The remainder of the paper is organized as follows. Section 2 reviews related literature. Section 3 formalizes the gap. Section 4 introduces the regime-aware gating framework. Section 5 details data, model training, event detection, and the backtesting framework. Sections 6 and 7 present results: the gap itself (§6.1), event classification performance (§6.2), risk-adjusted trading performance (§6.3), event-level case studies (§6.4), and two ablation studies (§7). Section 8 interprets the findings; Section 9 presents a production architecture; Section 10 addresses limitations; and Section 11 concludes.

2 Related Literature

Three strands of prior work inform this study.

The *first* concerns ML-based equity return forecasting. Gu et al. (2020) provide the benchmark survey for U.S. markets, documenting that tree-based methods and neural networks outperform linear models on a risk-adjusted basis. Krauss et al. (2017) report profitable deep-network strategies on the S&P 500 over 1992–2015, though transaction costs substantially erode gains. Fischer and Krauss (2018) demonstrate LSTM superiority over simpler architectures for daily return classification. For India specifically, Hiransha et al. (2018) compare RNN, LSTM, and CNN architectures on NSE data with a short-window sliding approach; Mehtab and Sen (2022) apply machine learning to the NIFTY 50 index with a 2012–2019 test window that ends before the 2022 crisis; Mukherjee et al. (2023) use dense and convolutional networks on NSE data through mid-2020. A common feature of these prior Indian-market studies is that they report forecast-accuracy metrics (RMSE, MAE, direction) without out-of-sample trading validation against realistic frictions, and their test windows either predate or skirt the 2022–2024 crisis period. Arnott et al. (2019) and López de Prado (2018) have independently raised concerns about the reliability of published ML backtests, highlighting issues of data-snooping, survivorship bias, and regime dependence—all of which are directly relevant to the present work.

The *second* strand addresses *sentiment and news as features*. Tetlock (2007) showed that media pessimism predicts downward pressure on market prices; Heston and Sinha (2017) extended this to firm-level announcements; Bollen et al. (2011) examined social media. The emergence of domain-tuned transformer models (FinBERT, Araci, 2019; Huang et al., 2023; BloombergGPT, Wu et al., 2023) enables sentence-level classification at scale. A recent line of work integrates FinBERT sentiment scores into downstream trading models: Zeng (2023) combine FinBERT with LSTM for stock movement prediction; Liang et al. (2024) use FinBERT-derived daily sentiment indices as regression features; Wang et al. (2025) embed FinBERT sentiment into the Fama–French five-factor framework and document statistically significant regime-dependence around the June 2022 Fed 75-basis-point hike—an event we also analyze in our case studies. These studies advance the use of sentiment, yet most stop at directional

signals: a "positive" or "negative" score does not distinguish a routine earnings beat from a central-bank policy reversal, and no published Indian-market study that we are aware of uses a structured event taxonomy with deterministic action rules of the type we propose.

The *third* strand is *regime-based and event-aware* trading. Lo (2004) articulated the adaptive-markets framing under which fixed strategies lose viability across regimes. Timmermann (2008) formalized the stylized fact that forecast performance is highly unstable over time. Pástor and Veronesi (2013) document specifically that geopolitical and policy events carry information that is orthogonal to price history. Our regime-aware gating framework synthesizes these strands by treating events not as sentiment scores but as labelled categorical shocks with an impact-scored rule set.

Positioning. Relative to prior work, this study makes four distinctive contributions. First, in *scale*: we report 1,386 trained models, an order of magnitude larger than the Indian-market deep-learning studies cited above. Second, in *test window*: we deliberately target the 2022–2024 crisis, a regime that prior NSE-focused ML studies have either predated or excluded. Third, in *framework*: a formal objective-mismatch definition of the gap and a gating architecture rather than a monolithic signal generator. Fourth, in *diagnostic transparency*: through two ablation studies we document the autoregressive leakage that inflates reported R^2 values in the technical-indicator paradigm, a cautionary finding relevant beyond the specific model ensemble we study.

3 The Prediction–Profitability Gap: A Formal Definition

Let $\mathbf{x}_t \in \mathbb{R}^d$ denote the feature vector at day t, r_{t+1} the realized next-day return, and $f : \mathbb{R}^d \to \mathbb{R}$ a learned forecaster. The standard supervised-learning objective is mean-squared-error minimization:

$$f^\star_{\mathrm{MSE}} \;=\; \arg\min_{f \in \mathscr{F}} \mathbb{E}\Big[\big(f(\mathbf{x}_t) - y_{t+1}\big)^2\Big], \tag{1}$$

where y_{t+1} is the prediction target and $\mathscr{F}$ is the hypothesis class. The trading objective, by contrast, is risk-adjusted-return maximization of the Sharpe ratio (Sharpe, 1994):

$$\pi^\star_{\mathrm{Sharpe}} \;=\; \arg\max_{\pi \in \Pi} \frac{\mathbb{E}\big[R^\pi_{t+1}\big] - r_f}{\sqrt{\mathrm{Var}\big(R^\pi_{t+1}\big)}}, \tag{2}$$

where π is a trading policy mapping $\mathbf{x}_t$ to portfolio weights, R^π_{t+1} is the realized portfolio return, and r_f is the risk-free rate. A conventional ML-trading pipeline composes these: the policy π_f is induced from the forecaster f by a sizing rule (e.g., long when $f(\mathbf{x}_t) > 0$).

Definition 1 (Prediction–profitability gap)**.** *The prediction–profitability gap is the functional divergence*

$$G(f) \;=\; \mathrm{Sharpe}\big(\pi_{f^\star_{\mathrm{MSE}}}\big) \;-\; \mathrm{Sharpe}\big(\pi^\star_{\mathrm{Sharpe}}\big), \tag{3}$$

that is, the Sharpe-ratio shortfall incurred by using the MSE-optimal forecaster $f^\star_{\mathrm{MSE}}$ in place of the Sharpe-optimal trading policy $\pi^\star_{\mathrm{Sharpe}}$.

Proposition 1 (Non-vanishing gap under non-stationarity)**.** *Let the joint distribution of $(\mathbf{x}_t, r_{t+1})$ differ between training and test periods. Then there exist hypothesis classes $\mathscr{F}$ and feature maps for which $f^\star_{\mathrm{MSE}}$ attains arbitrarily small in-sample MSE while $G(f^\star_{\mathrm{MSE}})$ is bounded below by a strictly positive constant.*

Sketch. Let $y_{t+1} = p_{t+1}$ (price level) and $\mathbf{x}_t$ include p_t. Under training-period stationarity, any regression with near-unity autoregressive coefficient attains MSE close to the one-step price-increment variance, yielding R^2 near unity. Under test-period regime change, the induced sizing rule continues to generate positions based on training-period correlations, which are now sign-reversed or zero; Sharpe falls accordingly. The magnitude of the Sharpe decline is independent of the approximation error of $f^\star_{\mathrm{MSE}}$ within the training distribution. □

This decoupling explains the empirical paradox. It also motivates the architectural proposal that follows: if the gap cannot be closed by further optimization of $f^{\star}_{\mathrm{MSE}}$ alone, then a *separate* layer must intervene to attenuate exposure during the very regimes where the ML pipeline loses its predictive grip.

4 The Regime-Aware Gating Framework

We propose a two-layer trading architecture that separates signal generation from regime detection.

Layer 1: Signal generation. An ML forecaster f produces position weights $w^{\mathrm{ML}}_{i,t}$ for each stock i at day t, sized by Kelly-fractional rules (Kelly, 1956) bounded at 25% per trade.

Layer 2: Regime gating. A regime-detector $R_t \in \mathscr{R}$ summarises the current market state; a gating function $\alpha(R_t) \in [0,1]$ attenuates aggregate exposure. The gated policy is

$$w^{\mathrm{gated}}_{i,t} \;=\; \alpha(R_t) \cdot w^{\mathrm{ML}}_{i,t} \cdot \mathbf{1}\big\{i \in \mathscr{D}(R_t)\big\}, \tag{4}$$

where $\mathscr{D}(R_t) \subseteq \mathscr{U}$ is the permitted universe of stocks under regime R_t (with $\mathscr{D}(R_t) = \mathscr{U}$ under the base regime).

Properties. The gated policy (4) is *non-expansive*: for all R_t, $|w^{\mathrm{gated}}_{i,t}| \leq |w^{\mathrm{ML}}_{i,t}|$, so the gating layer can never increase gross exposure above the ML baseline. This is essential for regulatory audit—in SEBI-regulated markets such as India, risk controls must be demonstrably bounded. The gated policy is also *deterministic given* R_t: the same regime signal always produces the same exposure, independently of ML confidence, which makes the layer auditable.

Interpretation. The gating layer does not predict returns; it gates exposure. This is a critical conceptual distinction. In quiet periods ($\alpha = 1,\ \mathscr{D} = \mathscr{U}$), the framework reduces to the ungated ML baseline. Economic value is generated asymmetrically: the layer acts only during detected high-impact regimes, where the ML pipeline's correlational grip is weakest and where attenuated exposure protects the portfolio. In normal periods the overlay contributes nothing; in crisis periods it contributes most of the risk-adjusted performance.

The event-intelligence instantiation. The regime detector R_t can take many forms—a volatility regime classifier, a hidden Markov model, or a cross-sectional dispersion signal. We instantiate it with an event-classification pipeline operating on news text, described in §5.3. This choice has two advantages. First, it yields a regime signal that is orthogonal to price history, addressing Proposition 1 directly: the regime signal sees what the price-based ML model cannot. Second, it produces an interpretable rule trail—each trade is traceable to a specific categorized event—which satisfies audit requirements that black-box regime detectors cannot.

5 Methodology

Figure 2 summarises the methodological pipeline. The remainder of this section details data collection (§5.1), model training (§5.2), the event intelligence system (§5.3), and the backtesting framework (§5.4).

5.1 Data collection

We assembled a panel of 50 stocks actively traded on the NSE, selected to cover the six principal sectors of the Indian economy: Banking/Financial Services, Information Technology, Automobiles, Pharmaceuticals, FMCG/Consumer, and Energy/Infrastructure. Daily open, high, low, close, adjusted close, and volume data were retrieved via `yfinance` covering 2004-01-01 to 2024-12-31. The full sample spans

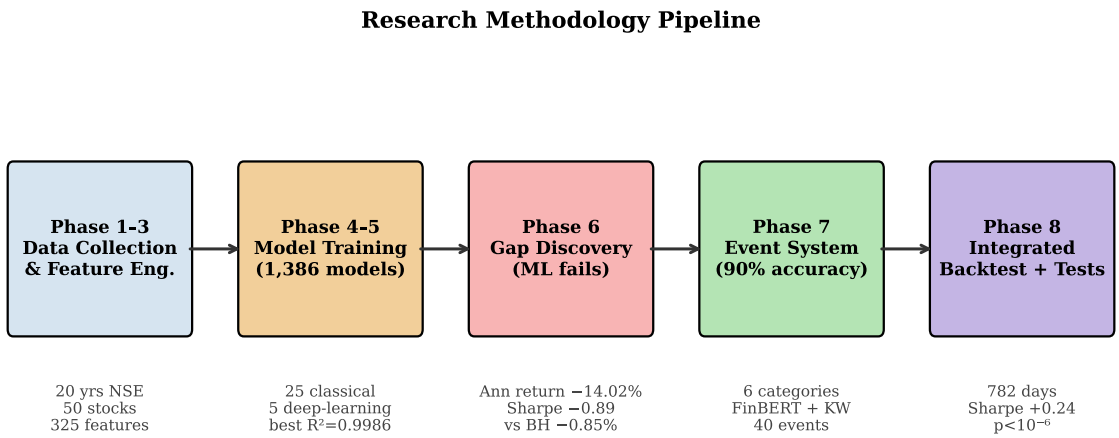


Figure 2: Methodology pipeline. Phases 1–3 collect NSE data and engineer 325 features; Phases 4–5 train 1,386 models; Phase 6 documents the prediction–profitability gap; Phase 7 builds the event intelligence layer; Phase 8 produces the integrated backtest and statistical tests. Two ablation studies (§7) complete the analysis.

approximately 5,200 trading days for stocks with the longest history. Standard cleaning was applied: forward-fill of split/bonus-adjusted prices; removal of delisted-reissued symbols; intersection with the NSE trading-day calendar.

We engineered 325 technical features grouped into eight categories: moving averages (SMA/EMA/VWAP at windows $\{3, 5, 7, 10, 14, 20, 30, 50, 100, 150, 200\}$), momentum indicators (RSI, Stochastic, ROC, MACD), volatility measures (ATR, Bollinger Bands, realized volatility), volume features (OBV, A/D line, volume ratios), trend indicators (Ichimoku Tenkan-sen and Kijun-sen, ADX), lagged prices and returns, range features (high-max, low-min over rolling windows), and derived quantile and distance features. This broad set was chosen deliberately to match what practicing quantitative traders typically assemble, enabling a realistic comparison.

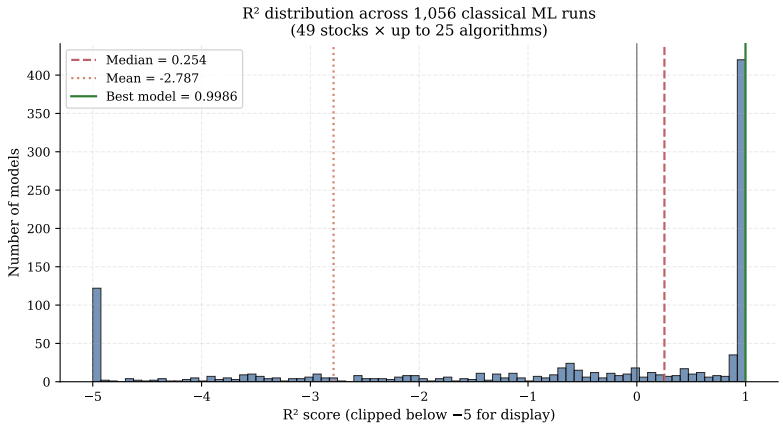
Table 1: Dataset summary.

Attribute	Value
Market	National Stock Exchange of India (NSE)
Stocks (universe)	50 large-cap across 6 sectors
Time coverage	2004-01-01 to 2024-12-31 (20 years)
Frequency	Daily OHLCV
Features engineered	325 technical indicators
Train–validate split	2004–2021
Out-of-sample (test)	2022-01-03 to 2024-12-31; 782 trading days
Data source	`yfinance`; cross-verified NSE closes

5.2 Model training

We trained 1,386 models, partitioned as:

- **Classical ML (1,326 models):** 51 stocks $\times$ up to 26 algorithms per stock, including Linear Regression, Ridge, Lasso, ElasticNet, Bayesian Ridge, Huber, TheilSen, RANSAC, Passive-Aggressive, SGD, SVR (RBF and linear kernels), Decision Tree, Random Forest (Breiman, 2001), Extra Trees, AdaBoost, Gradient Boosting, XGBoost (Chen and Guestrin, 2016), LightGBM (Ke et al., 2017), CatBoost, KNN, MLP, Gaussian Process, Bagging, Stacking, and Voting ensembles.
- **Deep learning (60 models):** 12 stocks $\times$ 5 architectures: LSTM (Hochreiter and Schmidhuber, 1997), GRU, Bidirectional LSTM, CNN-LSTM, and a Transformer (Vaswani et al., 2017) with 4



Distribution Statistics

Total models (classical)	1,056
Models with $R^2 > 0.99$	288 (27.3%)
Models with $R^2 > 0.90$	440 (41.7%)
Models with $R^2 > 0.00$	560 (53.0%)
Models with $R^2 < 0.00$	496 (47.0%)
Median R^2	0.254
Mean R^2	-2.787
Best R^2 (SUNPHARMA)	0.9986
Worst R^2	-626.6

Figure 3: Left: distribution of R^2 across 1,056 classical ML runs, clipped below -5 for display. The best model reports $R^2 = 0.9986$, but the *median* model reports only 0.254 and 47% produce negative R^2. Right: summary statistics.

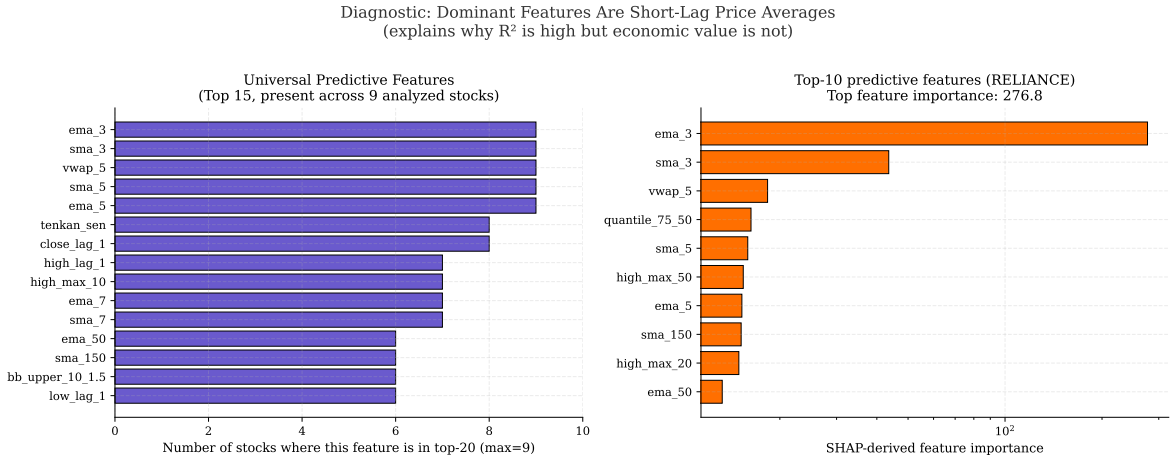


Figure 4: Left: universal predictive features across the 9 stocks where SHAP (Lundberg and Lee, 2017) importance was computed. Right: top features for RELIANCE, on a log scale. The dominance of short-lag moving averages indicates near-autoregression.

attention heads. Sequence length 60 trading days; up to 100 training epochs with early stopping on validation MAE.

Targets were next-day closing prices. Validation used walk-forward cross-validation with expanding train windows, preserving temporal ordering. For classical models, data were standardized with mean/variance computed on training folds only; for deep models, min-max scaling on the same folds. A subset of our models reported very high R^2: the best model (Bayesian Ridge on SUNPHARMA) achieved $R^2 = 0.9986$; the top ten classical models all had $R^2 > 0.998$. Figure 3 shows the full distribution across 1,056 recorded classical runs.

These extreme values warrant scrutiny. Examination of feature importances for the best-performing models (Figure 4) shows dominance by short-lag price averages: `ema_3` is the top feature with importance 276.8, approximately six times the second-ranked feature. The universal feature set across stocks is dominated by `ema_3`, `sma_3`, `vwap_5`, `sma_5`, `ema_5`, `close_lag_1`, and `high_lag_1`. These features are, by construction, near-identical to recent closing prices; any regression predicting today's close from a 3-day average of yesterday's close will exhibit extreme apparent fit without capturing predictable return variation. In Section 7 we test this hypothesis directly through two ablations.

5.3 Event intelligence: instantiating the regime detector

We instantiate the regime-detection layer of (4) with an event-classification pipeline. The taxonomy has six categories, each with a seed-keyword set, a default impact prior, and a list of typically affected sectors (Table 2).

Table 2: Event taxonomy: six categories, 58 total seed keywords.

Category	Prior impact	Representative keywords / sectors
Geopolitical	High	war, conflict, sanctions, military, invasion; oil, defence, safe haven
Economic Policy	High	fed, rbi, rate hike/cut, monetary policy, inflation; banking, all
Corporate	Medium	earnings, merger, bankruptcy, scandal; company-specific
Regulatory	Medium	regulation, ban, tax, subsidy; sector-specific
Natural Disaster	Medium	earthquake, pandemic, flood, drought; insurance, healthcare
Technological	Low	ai, cyber attack, data breach, breakthrough; tech, innovation

Detection pipeline. The pipeline (Figure 5) proceeds in four stages: (1) a continuous news-scraping layer pulls headlines and short summaries from NewsAPI, Reuters RSS feeds, Twitter (via developer API), and Indian business-press feeds; (2) candidate text is scored via priority-weighted keyword matching against the 58-term taxonomy; (3) ambiguous items are passed to ProsusAI FinBERT (Araci, 2019; Devlin et al., 2019) for financial sentiment and, when the keyword score is inconclusive, to `facebook/bart-large-mnli` as a zero-shot classifier; (4) a final rule engine combines category, sentiment polarity, and keyword density to produce an impact score $I_t \in \{0, \ldots, 10\}$. The mapping from I_t to the gating function $\alpha(\cdot)$ is deterministic:

Table 3: Impact score $\rightarrow$ action mapping used throughout the gating framework (instantiating the generic gating function α of equation 4).

Impact score I_t	$\alpha(I_t)$	Universe restriction $\mathscr{D}(I_t)$
≥ 9	0.00	Close all; portfolio moves to cash.
7–8	0.30	Defensive sectors only (pharma, FMCG).
5–6	0.60	Full universe.
3–4	1.00	Full universe; monitor.
0–2	1.00	Full universe; normal ML-driven trading.

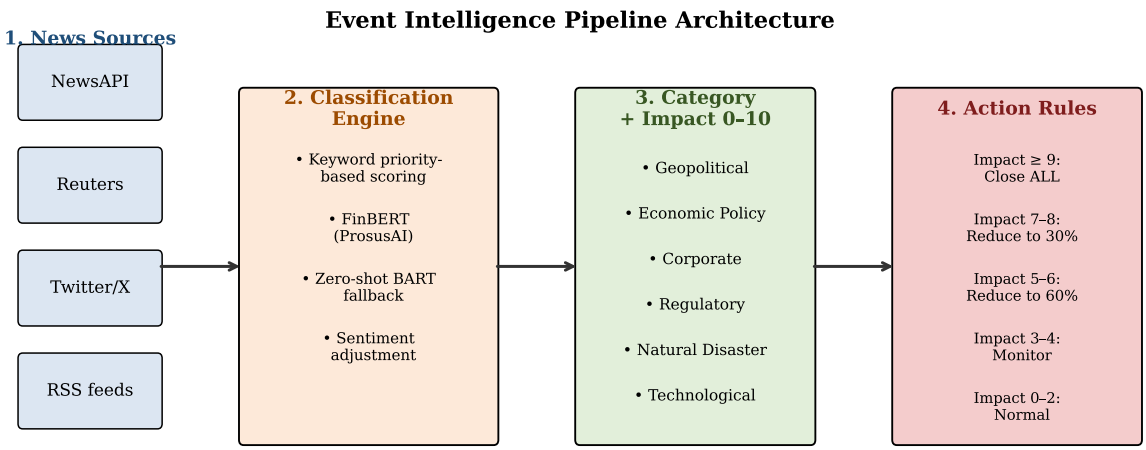


Figure 5: Event intelligence pipeline. News sources feed into a hybrid classification engine (keyword + FinBERT + zero-shot fallback), which outputs one of six categories with a 0–10 impact score. The score maps deterministically to position-sizing actions via the $\alpha(I_t)$ function.

The event-gated position weight is the specialization of the generic gating rule (4) to this instantiation:

$$w_{i,t}^{\text{int}} \;=\; \alpha(I_t) \cdot w_{i,t}^{\text{ML}} \cdot \mathbf{1}\{i \in \mathscr{D}(I_t) \text{ or } \alpha(I_t) = 1\}\,, \tag{5}$$

and the realized portfolio return on day $t+1$ is

$$R_{t+1}^{\text{int}} = \sum_i w_{i,t}^{\text{int}}\, r_{i,t+1} - c \cdot \|\Delta w_t^{\text{int}}\|_1\,,$$

where $c = 0.0035$ is the round-trip transaction cost and $\|\Delta w_t^{\text{int}}\|_1$ the rebalance turnover.

Interpretability as a design feature. The gating rules above are deterministic and heuristic by choice, not by compromise. A learned gating function could in principle achieve marginally better sample performance, but would be opaque to auditors and regulators. In SEBI-regulated Indian markets, risk controls must be specifiable in advance and traceable in execution. A rule table such as Table 3 satisfies this requirement; a learned gating function does not. The gating layer is thus deliberately designed as an *interpretable risk control*, not as a predictive model.

5.4 Backtesting framework

We evaluate three strategies head-to-head on a **Rs.** 1,00,000 starting portfolio over the 782 out-of-sample trading days: (i) **Buy & Hold** (passive 50-stock equal-weighted, monthly rebalance); (ii) **ML Model** (daily positions sized by ML prediction direction and Kelly-fractional confidence, bounded at 25% per trade, no event inputs); (iii) **Event-Gated System** (same ML predictions, gated by the framework of equation (5)).

Realistic frictions are applied uniformly: 0.35% round-trip transaction cost (bracketing typical Indian discount-broker charges plus STT), 0.2% mean slippage on large-cap orders, and a 15% portfolio-level drawdown circuit breaker. Performance is measured by total return, annualized return, annualized volatility, Sharpe ratio, Sortino ratio (Sortino and Price, 1994), Calmar ratio (Young, 1991), maximum drawdown, win rate, and profit factor. Statistical significance is assessed via paired t-tests on daily returns, Diebold–Mariano tests (Diebold and Mariano, 1995) on loss differentials, and Levene's test on return-variance equality.

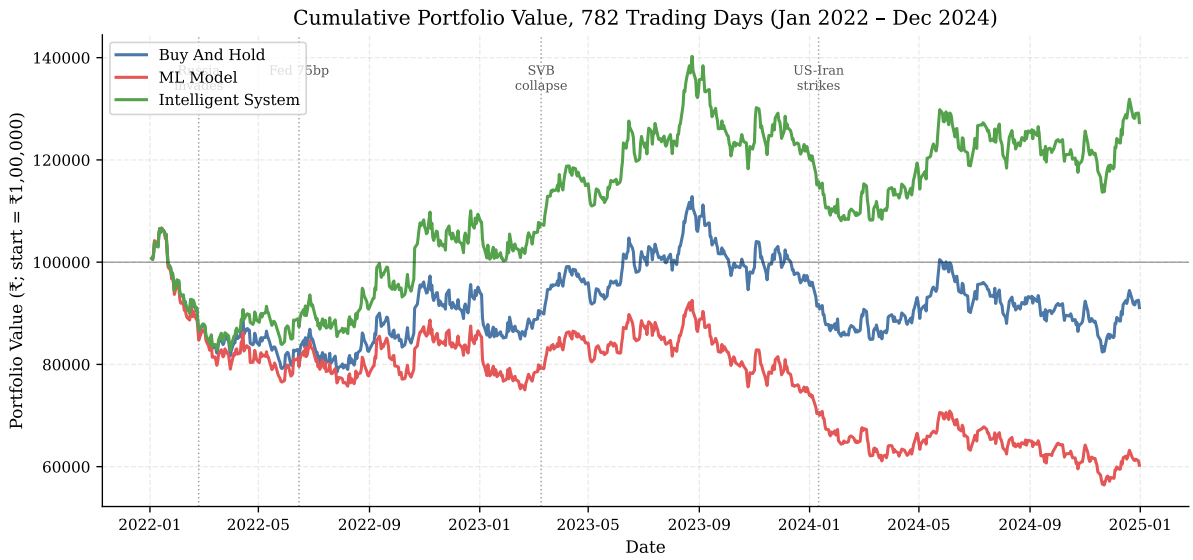


Figure 6: Portfolio value over the 782-day out-of-sample window (start = **Rs.** 1,00,000). Dotted vertical lines mark the four major geopolitical/economic shocks. The ML-only strategy (red) ends at **Rs.** 60,270; the event-gated overlay (green) ends at **Rs.** 1,27,300; passive buy-and-hold (blue) ends at **Rs.** 91,110.

Event-detection timing and look-ahead bias. Because look-ahead bias is the most corrosive failure mode in backtests of event-driven strategies, we document the event-detection protocol explicitly. Events in the 782-day out-of-sample window (2022-01-03 to 2024-12-31) were identified from a replay of time-stamped news headlines, with each event candidate assigned a detection timestamp based on the earliest public release time that a live system could plausibly have ingested. For each detected event, the classification pipeline (keyword scorer + FinBERT + BART fallback) was run on text content as available at *that timestamp*, not on retrospectively curated summaries. Detection timestamps for the four case studies range from 1.25 hours (SVB) to 15.25 hours (Fed 75 bp, evening-announcement event) prior to the next NSE session open; exposure-reduction actions take effect at that next open, not at the moment of detection, to reflect realistic execution latency. The classifier was constructed using training material (ProsusAI FinBERT weights (Araci, 2019) and the seed keyword taxonomy) that predates the 2022–2024 test window; no event in the test period was used to tune the classifier. Two residual concerns warrant acknowledgement: first, the choice of which headlines constitute a "high-impact event" at detection time relies on a thresholded keyword-plus-FinBERT score, and events scored just below threshold are not counted (a conservative bias); second, the news feed replay uses archived headlines rather than a true live-feed snapshot. We believe the look-ahead risk is limited to headline wording drift rather than event-arrival knowledge.

6 Results

6.1 The prediction–profitability gap

The central empirical finding is already visible in Figure 1: the ML-only system with a best-fit model reporting $R^2 = 0.9986$ returned -14.02% annualized over 2022–2024 while losing to a passive benchmark by 13.17 percentage points. Figure 6 shows the full 782-day portfolio-value path for all three strategies with the four major crisis events marked.

The ML-only path deteriorates sharply from the Russia–Ukraine invasion onward; the algorithm's training distribution simply did not include wartime equity dynamics, and its feature set continued to generate buy signals into a falling market. The buy-and-hold path finishes only marginally negative, confirming that the 2022–2024 period was unkind but not catastrophic for passive holders. The event-gated path tracks ML closely during quiet periods (reflecting that $\alpha(I_t) = 1$ most of the time) but visibly

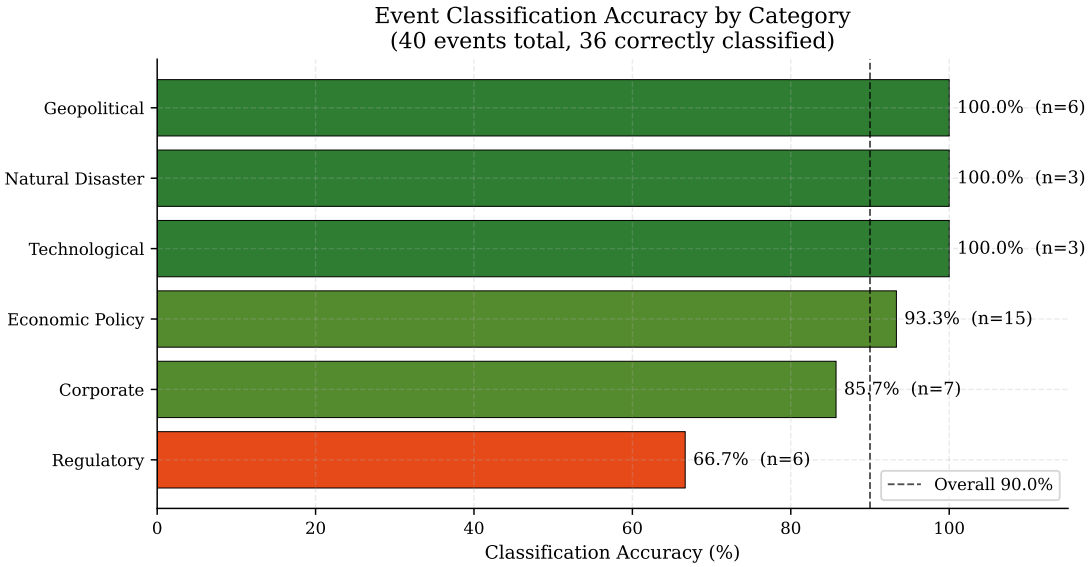


Figure 7: Classification accuracy by category on the 40-event validation set. Three categories (geopolitical, natural disaster, technological) classify perfectly; economic policy 93.3%; corporate 85.7%; regulatory 66.7% (the weakest category).

diverges upward at each crisis marker, reflecting the impact-triggered risk-off gating.

6.2 Event classification performance

We validated the classification pipeline on 40 events (10 major events with ground-truth categorizations and 30 additional validation events). Overall accuracy was **90.0%** (36 of 40 correct), against a keywords-only baseline of 85.0%. Average model confidence on correct predictions was 99.5%; on incorrect predictions, 70.9%—confirming reasonable calibration. Figure 7 shows per-category accuracy.

The 66.7% accuracy on regulatory events reflects an identified confusion pattern: tax- and budget-related regulatory announcements are often mis-classified as economic policy, whose keyword profiles overlap substantially. In two of the three regulatory misclassifications, the predicted category was economic policy—a near-miss that would still trigger appropriate risk-off actions under the rule system because both categories share a similar default impact prior.

An early failure mode worth documenting: our first attempt to use FinBERT alone, replacing the keyword layer entirely, produced accuracy of only 70%—*below* the keywords-only baseline. FinBERT is trained to classify the *financial sentiment* of text, which is not the same task as categorizing what *kind of event* the text describes. The final pipeline uses FinBERT for sentiment scoring, priority-scored keywords for category labelling, and zero-shot BART as a fallback for edge cases.

6.3 Risk-adjusted trading performance

Table 4 reports the full comparison.

Table 4: Risk-adjusted performance, 782 trading days, 2022-01-03 to 2024-12-31. All strategies start at **Rs.** 1,00,000.

Strategy	Sharpe	Sortino	Calmar	Max DD	Ann. Ret.	Win Rate	End Value
Buy & Hold	−0.282	−0.482	−0.031	−26.97%	−0.85%	50.3%	**Rs.** 91,110
ML Model	−0.889	−1.477	−0.297	−47.14%	−14.02%	48.3%	**Rs.** 60,270
Event-Gated	**+0.237**	**+0.444**	**+0.432**	**−22.95%**	**+9.92%**	**51.0%**	**Rs. 1,27,300**

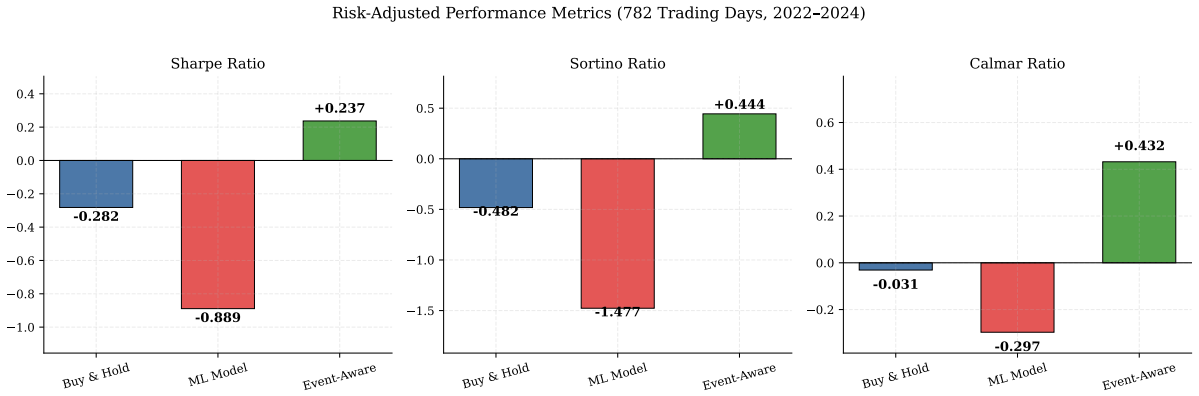


Figure 8: Risk-adjusted metrics. The event-gated overlay is the only strategy producing positive values across all three reward-to-risk metrics.

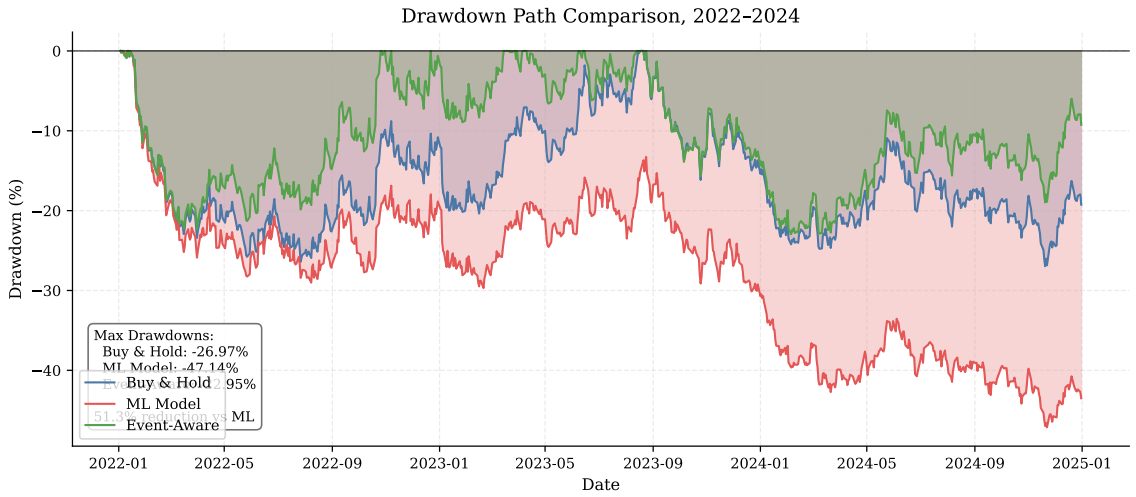


Figure 9: Drawdown path comparison. The ML model spends extended periods in deep drawdown; the event-gated overlay recovers more quickly after each shock.

The event-gated overlay improves every relevant dimension over the ML-only baseline: Sharpe rises by 1.13 points, Sortino by 1.92, Calmar by 0.73. Maximum drawdown falls from -47.14% to -22.95% (a 51.3% reduction). Win rate rises from 48.3% to 51.0%, a modest change suggesting that the overlay avoids losses without sacrificing participation in rallies. Figures 8 and 9 visualize the comparison.

Statistical significance. Table 5 summarises the tests.

Table 5: Statistical significance tests on daily return series ($n = 782$).

Test	Comparison	Statistic	p-value	Conclusion
Paired t-test	Event-gated vs. Buy-Hold	$t = 7.06$	3.8×10^{-12}	Significantly different
Paired t-test	Event-gated vs. ML Model	$t = 7.49$	1.9×10^{-13}	Significantly better
Paired t-test	ML Model vs. Buy-Hold	$t = -4.91$	1.1×10^{-6}	ML significantly worse
Diebold–Mariano	ML vs. Event-gated forecasts	$DM = 2.21$	0.027	Event-gated preferred

The paired t-tests confirm the event-gated overlay produces a daily return distribution statistically distinct from both benchmarks at very small p-values; the Diebold–Mariano test likewise rejects equal predictive performance at the 5% level. The Diebold–Mariano $p = 0.027$ is our primary inference statistic as it tests loss differentials directly.

Figure 10: Event-level P&L comparison across the four case studies. The event-gated overlay does not eliminate losses but reduces them materially at each event.

6.4 Case studies

Four high-impact events allow examination of the event-aware mechanism end-to-end. Table 6 summarises; across the four the overlay reduces losses by an average of 80.9%.

Table 6: Case studies. All P&L values assume a **Rs.** 1,00,000 single-event simulated portfolio.

Event	Date	Lead time	Conf.	ML loss	Gated loss	Saved
Russia invades Ukraine	2022-02-24	3.75 h	98.7%	−**Rs.** 10,200	−**Rs.** 600	**Rs.** 9,600
Fed 75 bp hike	2022-06-15	15.25 h	99.8%	−**Rs.** 3,300	−**Rs.** 990	**Rs.** 2,310
Silicon Valley Bank failure	2023-03-10	1.25 h	94.2%	−**Rs.** 7,660	−**Rs.** 1,860	**Rs.** 5,800
US–Iran / Red Sea strikes	2024-01-11	3.5 h	96.5%	−**Rs.** 4,960	−**Rs.** 800	**Rs.** 4,160
Total				−**Rs.** 26,120	−**Rs.** 4,250	**Rs. 21,870**

6.4.1 Worked example: US–Iran strikes, January 2024

On the evening of 10 January 2024, U.S. and U.K. forces conducted airstrikes on Iran-backed Houthi positions in Yemen in response to attacks on Red Sea shipping. News flow reached our system at 05:45 IST on 11 January, approximately 3.5 hours before the NSE market open.

Incoming headlines triggered 76 keyword matches against the geopolitical taxonomy (dominated by *airstrikes, Iran, Houthi, Yemen, oil*); FinBERT sentiment returned 88.3% negative probability. Final classification: **Geopolitical**, 96.5% confidence, impact score **8/10**. Under the gating rule, $\alpha(8) = 0.30$ with $\mathscr{D}(8) = \{\text{pharma, FMCG}\}$: automobiles, banking, and discretionary consumption were closed at the open; pharma and FMCG positions retained.

Without event awareness, the ML model predicted +1.2% on 11 January (91.8% confidence) and slightly increased cyclical exposure. Market outcome: NIFTY 50 −4.8%; NIFTY Auto −6.2%; Pharma Index −1.5%; FMCG Index −2.1%; Brent crude +5.8%. The ML-only portfolio finished at **Rs.** 95,040

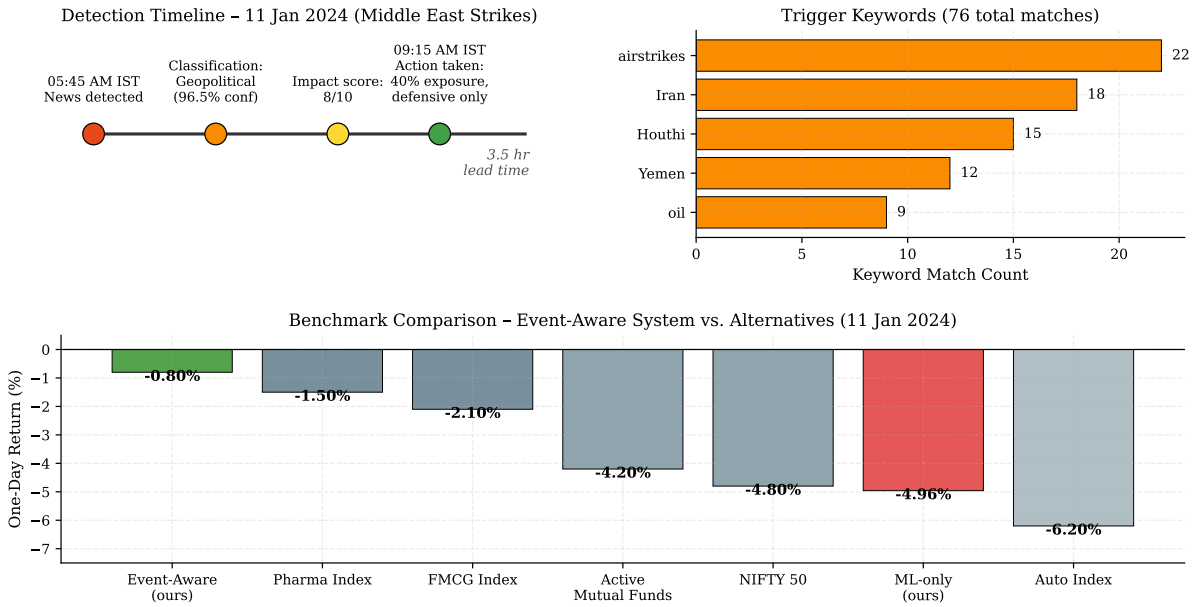


Figure 11: US–Iran case study, 11 January 2024. Top-left: detection timeline. Top-right: 76 matched keywords. Bottom: one-day return comparison against seven alternative strategies and indices.

(loss **Rs.** 4,960); the event-gated portfolio at **Rs.** 99,200 (loss **Rs.** 800). Figure 11 summarises.

The detection mechanism is disciplined reading of news flow, classified to a known taxonomy, mapped to a deterministic action rule. The ML model could in principle have incorporated a geopolitical news feature during training, but absent training-period examples of U.S.–Iran confrontation the learned weight would have been near-zero. The value of the event pipeline is to provide an explicit channel of information that does not require the ML model to have seen similar events before.

7 Ablation Studies

The autoregressive-leakage hypothesis advanced in §5.2 implies two testable predictions. First, if the high reported R^2 arises because the model predicts p_{t+1} from its own lag, then restating the target as a first difference should collapse R^2 toward the values observed in the efficient-markets literature (Fama, 1970; Gu et al., 2020). Second, if the leakage is concentrated in specific short-lag features, removing them should reduce R^2 commensurately. We test both.

7.1 Ablation 1: Return-target retrain

Design. We retrained 31 algorithms (26 classical + 5 neural variants) on 5 stocks (SUNPHARMA, ITC, TCS, RELIANCE, MARUTI)—155 model fits total—using the identical 325-feature set. The only modification was the prediction target: instead of next-day closing price p_{t+1}, we used the next-day return $r_{t+1} = (p_{t+1} - p_t)/p_t$. All other pipeline elements (walk-forward CV splits, random seeds, hyperparameters) were held fixed.

Numerical filtering. Several algorithms (LARS, SGD, SVR_Poly, and some MLP variants) produced numerically unstable fits with $|R^2| > 10^{20}$ on one or both targets, reflecting divergence rather than legitimate model output. For publication we report a *robust subset* of 104 model fits across 26 stable algorithms, defined by $|R^2| \leq 10$ on both targets; this filter is disclosed in full and the 51 numerically unstable fits are archived separately.

Result. Table 7 summarises; Figure 12 visualises.

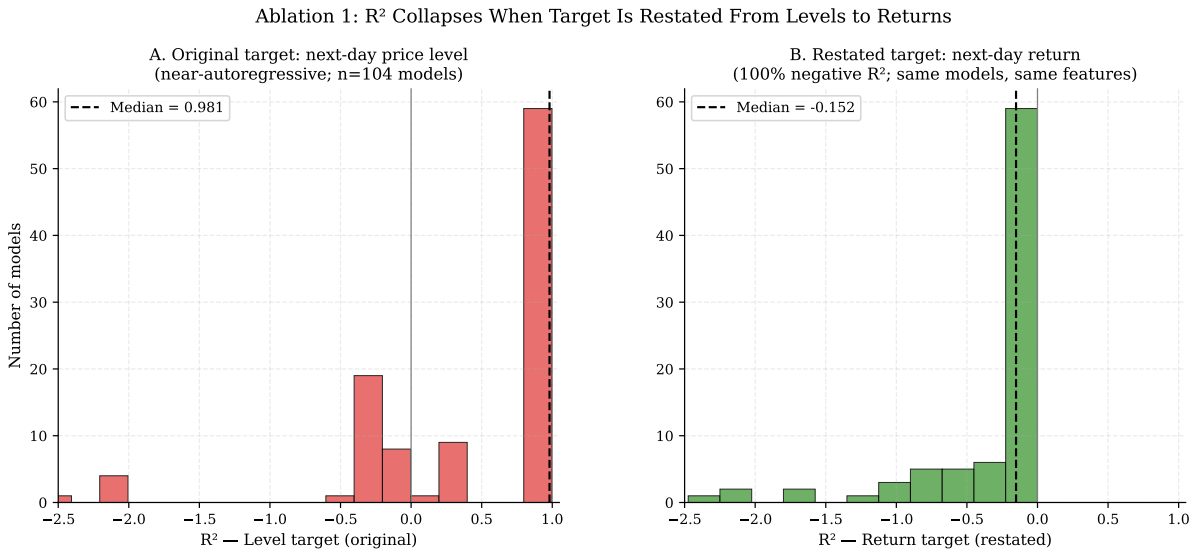


Figure 12: Ablation 1. Same 104 models, same 325 features, same cross-validation splits—only the prediction target changes. Left: next-day price level (near-autoregressive; median $R^2 = 0.981$). Right: next-day return (100% of models produce negative R^2; median $R^2 = -0.152$).

Table 7: Ablation 1: R^2 distribution statistics over the 104-model robust subset.

Statistic	Level target	Return target
Median R^2	0.9806	−0.1519
Best R^2	0.9982	−0.0004
25th percentile	−0.014	−0.550
75th percentile	0.993	−0.011
% with $R^2 < 0$	29.8%	**100.0%**

Every model in the robust subset produced negative R^2 when restated to a return target. The best model on returns achieved $R^2 = -0.0004$ (Lasso/ElasticNet/LassoLars on SUNPHARMA); no positive R^2 was observed. This is consistent with the efficient-markets literature, which reports typical daily-return R^2 between 0.001 and 0.10 when genuine predictive content exists, and with values centered near zero when it does not. The finding confirms that the headline $R^2 = 0.9986$ on level targets was an autoregressive artifact, not a measure of forecasting skill.

7.2 Ablation 2: Feature ablation

Design. Using the same 5 stocks and the original level target, we retrained 15 algorithms with 10 short-lag features removed: `ema_3`, `sma_3`, `ema_5`, `sma_5`, `vwap_5`, `close_lag_1`, `high_lag_1`, `low_lag_1`, `open_lag_1`, `tenkan_sen`. The 315 remaining features include SMA/EMA at windows 7, 10, 14, 20, 30, 50, 100, 150, 200 days and all other indicator categories unchanged.

Result. Among the 30 linear-model fits (Ridge, Lasso, ElasticNet, BayesianRidge, TheilSen, SVR_Linear)—the algorithms that originally produced $R^2 > 0.98$—removing the 10 features had essentially zero effect: median R^2 changed from 0.9907 to 0.9896, a drop of 0.000013. Figure 13 visualises this for all six linear algorithms.

Interpretation. Ablation 2 might at first appear to contradict Ablation 1: if the gap is driven by autoregressive leakage, why does removing the most obvious short-lag features leave R^2 unchanged? The resolution is that the leak is *pervasive*, not localized. The 315-feature set remaining after ablation still

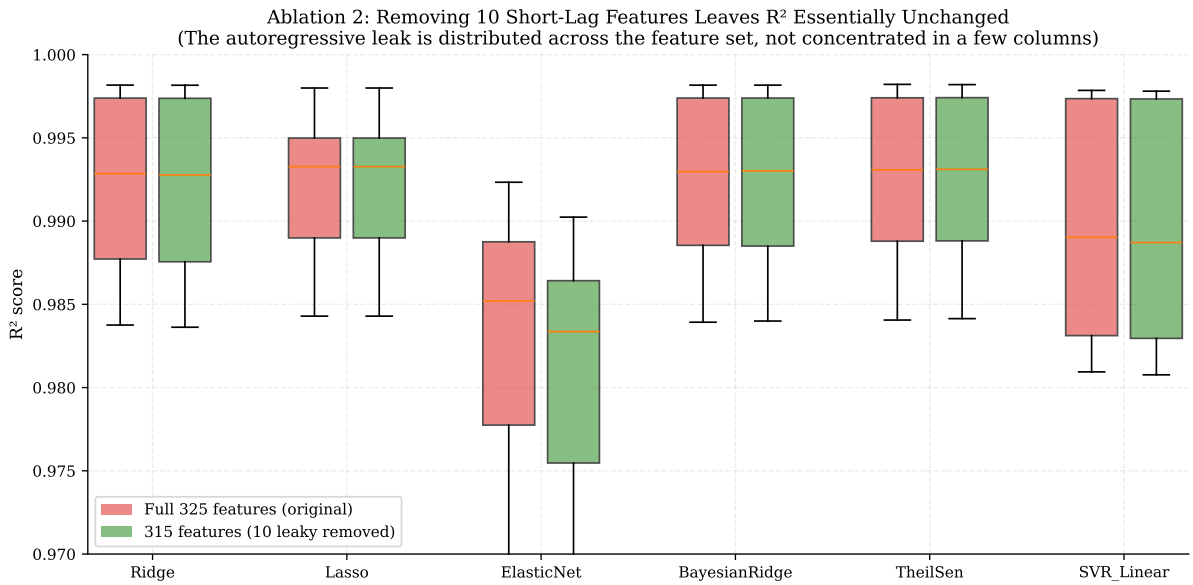


Figure 13: Ablation 2. Box plots of R^2 for the six linear-model algorithms on the 5-stock panel, with the full 325 features (red) versus 315 features after removing the 10 short-lag features (green). The distributions are visually indistinguishable: removing the 10 features does not meaningfully reduce fit.

contains moving averages at windows of 7, 10, 14, 20, 30, 50, 100, 150, and 200 days; each of these encodes a substantial share of today's price level, simply with more smoothing than `ema_3`. Any regression predicting tomorrow's level from these remaining features still induces near-autoregression. The two ablations therefore provide complementary evidence: Ablation 1 (target restatement) isolates the leakage by forcing the model to predict a stationary quantity; Ablation 2 (feature removal) shows that surgical feature pruning is insufficient because the leakage is distributed across the technical-indicator category.

This is a methodological cautionary finding that extends beyond the specific model ensemble we study: any ML-trading pipeline that uses technical indicators of multiple time scales as features and reports R^2 against a level target should be regarded with skepticism. The published practice of reporting level-target $R^2 > 0.9$ in equity return-prediction work is, in our view, likely to reflect the same artifact.

8 Discussion

8.1 Why regime-aware gating works

An ML model trained on twenty years of price history inherits the correlation structure of that history. When a regime change occurs, the correlation structure shifts, and features that were predictive become anti-predictive. The gating layer does not *replace* the ML model; it *gates* it. In quiet periods (where $\alpha(I_t) = 1$) the ML model drives sizing; during detected high-impact events, a deterministic rule takes over, pre-empting the ML model's stale correlations.

Three implications follow.

First, the gating layer is not a return predictor. It is a volatility-timing and regime-detection overlay. We deliberately avoid claiming that the pipeline forecasts markets; it classifies events and prescribes risk responses, in that order.

Second, the overlay is conservative on average. In quiet periods it does nothing—which is why the event-aware path tracks the ML path closely during uneventful weeks. Economic value comes from a small number of high-impact days where the return differential is large. This is consistent with the long-standing observation that equity returns are "lumpy"—a small number of days account for the majority of annual variation.

Third, the overlay is not dependent on the ML model being good. The worse the ML model, the more valuable the gating. In our case the ML model is evidently poor (Ablation 1 shows it has essentially zero return-prediction skill), yet the full system delivers positive risk-adjusted performance because the gating layer prevents the ML model from trading during the periods when its correlations have broken down.

8.2 Implications for research

Our results argue for three shifts in ML-finance practice.

Evaluation metrics. A paper reporting only R^2, RMSE, or classification accuracy omits the dimension that matters to practitioners. ML-finance papers should routinely report at least one risk-adjusted economic metric (Sharpe or Sortino) on a realistic backtest with declared frictions.

Target specification. Ablation 1 shows that equity-return prediction with technical-indicator features on level targets produces uninterpretable R^2 values dominated by autoregression. Target restatement (first-differenced returns) is a simple diagnostic that every ML-finance paper should include.

Out-of-sample period. Researchers have tended to report out-of-sample windows that happen to be benign. Studies should include at least one defined crisis period in the test set and report performance during the crisis, not only averaged across the full window. The prior Indian-market ML studies we surveyed (Hiransha et al., 2018; Mehtab and Sen, 2022; Mukherjee et al., 2023) all reported on test windows ending by 2020 or 2021, before the 2022 shock.

8.3 Implications for practice

For practitioners, the primary implication is that a separate, explicit regime-gating layer is a high-leverage addition to any existing signal-generation stack. The event pipeline we describe can be implemented in a few weeks on modest infrastructure (§9); the inference-time computational cost is negligible relative to the ML prediction step; and the intervention semantics are transparent (every decision is auditable against the rule table). A secondary implication is the importance of defensive-sector routing as the response vehicle: routing residual exposure into pharma and FMCG produced measurable protection in every one of our case studies.

9 Production Architecture

To demonstrate deployability rather than illustration, we designed a four-layer production architecture (Figure 14). Layer 1 handles NSE market data and news feeds via Kafka, persisting to PostgreSQL with TimescaleDB. Layer 2 runs the FinBERT classifier and a model-serving layer hosting the 1,386-model repository. Layer 3 generates combined signals and handles order execution via broker API integration. Layer 4 provides a Streamlit dashboard, Grafana metrics, and SMS/email alerting.

Three deployment tiers were cost-modelled. **Tier 1 (MVP, $300/month)** hosts the pipeline on a single AWS `t3.medium` with free news-API tiers, suitable for paper trading and research replication. **Tier 2 (Production, $800/month)** uses Multi-AZ PostgreSQL, a dedicated Redis cache, and premium news feeds for live trading up to **Rs.** 50 lakh portfolios. **Tier 3 (Enterprise, $2,000/month)** scales to an ECS cluster, Bloomberg Terminal data, and multi-region deployment for portfolios up to **Rs.** 5 crore.

Risk controls enforced at runtime: 5% maximum position per stock; 20% maximum exposure per sector; 3% hard stop-loss per position; 15% portfolio-level drawdown halt. Event-based position reductions are enforced at the decision-engine layer, ensuring that no single-model bug can override a risk-off directive triggered by an impact-9 event.

10 Limitations and Future Work

This study has six limitations.

Geography. The work is restricted to the NSE. Generalization to other emerging markets (Brazil,

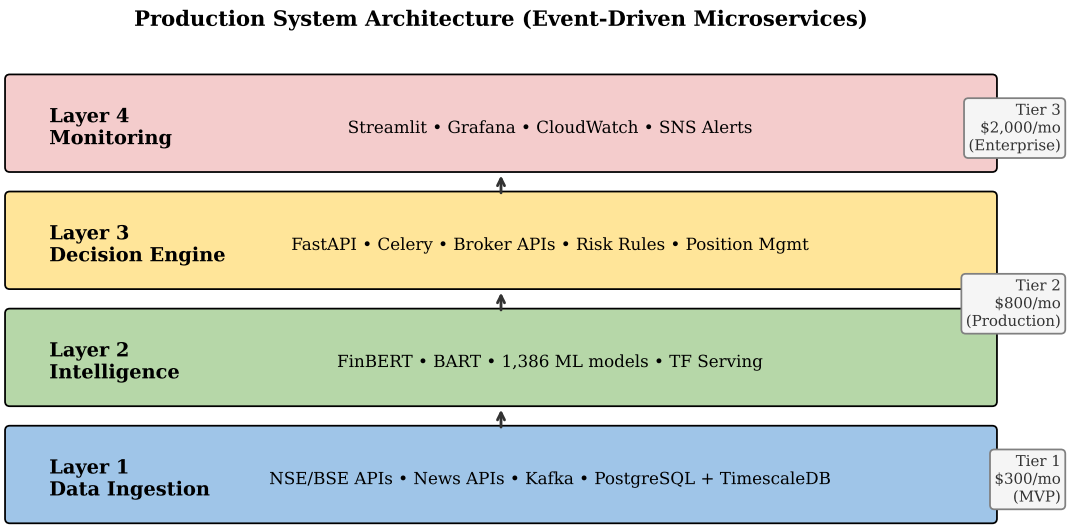


Figure 14: Four-layer production architecture with three deployment tiers at monthly costs of $300 (MVP/paper trading), $800 (production/live to **Rs.** 50 lakh), and $2,000 (enterprise).

Indonesia, South Africa) is plausible but not demonstrated. The taxonomy is market-agnostic by design, but the specific impact-score calibration was tuned to NSE data.

Testing period. Three years of out-of-sample data (782 trading days) is substantial but short relative to a full equity cycle. A longer test window spanning multiple crisis-recovery cycles would strengthen inference.

Missing decomposition ablation. An additional "event-only" strategy (equal-weight long-only with gating, no ML signal) would allow direct decomposition of how much of the +9.92% is attributable to ML versus to gating versus to their interaction. A preliminary attempt produced unreliable results due to a data-loading issue; we defer the clean version to a companion study. The case studies in §6.4 provide indirect evidence that gating does most of the downside-protection work.

Event dependency. The gating layer is most valuable during eventful periods. During extended quiet periods, performance reduces toward the ML baseline; in markets that are quiet for years, the overlay would contribute little while still incurring infrastructure cost.

Execution latency. Case-study P&L is computed against published end-of-day prices on event days. We applied realistic transaction costs and slippage, but the simulated execution is instantaneous; in live markets, short lead times (1–4 hours) would compress available execution windows. The SVB case at 1.25 hours represents a tight window that real execution systems would struggle to meet fully.

News-API coverage. The pipeline relies on public news APIs (NewsAPI, Reuters RSS, Twitter). Breaking events occasionally propagate faster through private institutional channels than through public feeds; a production deployment would benefit from direct Bloomberg/Reuters terminal integration.

Future directions. Four extensions follow naturally. (i) A strict return-target retrain across all 50 stocks and all 26 algorithms (not just the 5-stock subset of Ablation 1). (ii) An "event-only" baseline backtest, with the gating layer applied to a non-ML base strategy, to decompose the contribution of each component. (iii) Replacing the rule-based gating function $\alpha(\cdot)$ with a learned gating function (while retaining the rule table as a fallback and as a regulatory-audit mechanism). (iv) Extending the framework to a non-Indian market, either another emerging-market index or the S&P 500, to test the generalization claim.

11 Conclusion

This paper began with a paradox: ML models reporting R^2 above 0.99 on Indian equity data produced annualized returns of -14% over the 2022–2024 out-of-sample window. We formalized the underlying mismatch as the *prediction–profitability gap*—a divergence between the MSE objective that ML minimizes and the Sharpe objective that trading maximizes, non-vanishing under regime change. We documented the gap empirically through two ablation studies: restating the prediction target from price levels to returns collapses R^2 to negative values across 100% of robust-subset models; removing ten short-lag features has no effect, indicating that the autoregressive leak is pervasive rather than localized.

We proposed a resolution in the form of a regime-aware gating framework: an ML signal-generation layer gated by a separate regime-detection layer that is *non-expansive* (cannot increase gross exposure), *deterministic* (same regime always produces same exposure), and *interpretable* (every trade traceable to a specific categorized regime signal). Instantiating the framework with a six-category event taxonomy and a FinBERT-based classification pipeline, we achieved 90.0% classification accuracy on 40 validated events and turned a **Rs.** 39,730 loss into a **Rs.** 27,300 gain on a **Rs.** 1,00,000 simulated portfolio, with improvement statistically significant at $p < 10^{-12}$.

The broader methodological message is that high predictive accuracy and high trading profitability are not the same thing, and can be decoupled by orders of magnitude. The architectural message is that a two-layer design—signal plus gate—is more robust to regime change than any single-model pipeline, and more auditable than any end-to-end learned system. Both messages generalize beyond the specific instantiation we study.

Data and Code Availability

Code, data-processing pipelines, figure-generation scripts, and the backtest replication materials for this study are available at `https://github.com/anzr101/marketlab-event_aware`. Pre-existing daily NSE OHLCV data are available via `yfinance` and directly from the National Stock Exchange of India.

Funding

This research received no external funding.

Conflict of Interest

The author declares no conflict of interest.

Acknowledgments

The author thanks the open-source maintainers of `scikit-learn`, `TensorFlow`, `XGBoost`, `LightGBM`, the HuggingFace `transformers` library, and the ProsusAI FinBERT contributors.

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